

Ommkar Pattnaik

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Summary

Computer Science undergraduate with experience in AI agents, machine learning, and full-stack web development. Skilled in Python, JavaScript, and modern machine learning tools (TensorFlow, Scikit-learn) with a strong interest in building intelligent systems and data-driven applications. Passionate about Artificial Intelligence and developing scalable AI solutions. Certified in AI Agent Development and Data Analytics through IBM SkillsBuild.

Projects

AI-Powered Fitness Agent

Gemini API, Python, ReactJS

- Engineered a FastAPI backend with fully validated endpoints to generate AI-driven fitness programs using Gemini 2.5 Flash, providing responses within 1 minute.
- Designed structured prompt engineering techniques to produce tailored calorie targets, macros, workouts, and Indian-friendly meal plans.
- Implemented secure environment handling, CORS middleware, and clean REST architecture to enable seamless integration with a ReactJS frontend.
- Reduced manual fitness planning time by automating end-to-end plan generation, improving user experience through dynamic, real-time AI responses.
- Designed 150+ token structured prompt achieving consistent JSON output across 20+ test runs.

Malware Image Classification using MobileNetV2 Transfer Learning

MobileNetV2, CNN

- Built a 25-class malware family classifier using transfer learning on the MalImg dataset (~9,000+ binary visualization images), achieving 95% overall accuracy on the held-out validation set.
- Engineered a two-phase training pipeline: froze MobileNetV2 base to train classification head (94.3% val accuracy), then unfroze top 30 layers for fine-tuning at 1e-5 learning rate to prevent catastrophic forgetting.
- Converted grayscale malware binary images to 3-channel 96×96 RGB tensors for compatibility with ImageNet-pretrained MobileNetV2, enabling domain adaptation to cybersecurity visualization data.

Pokedex Web Application

HTML5, RESTful API, CSS

- Built an interactive single-page web application enabling users to search over 1000s of Pokémon by name, dynamically fetching real-time data from the PokéAPI.
- Implemented clean, responsive UI with efficient asynchronous JavaScript (Fetch API) to minimize latency and ensure smooth interactions.

Certifications

- Data Analytics Internship Certification – IBM SkillsBuild
- AI Agent Summer Internship Certification – IBM SkillsBuild

Education

B.Tech in Computer Science and Engineering

Odisha University of Technology and Research
CGPA: 8.93 (6th Semester)

2023 - 2027 (expected)

Skills

Languages: C, C++, SQL, JavaScript, Python

Web Development: HTML5, CSS3, ReactJS, REST APIs

Machine Learning & Data Science: TensorFlow, Scikit-learn, Pandas, NumPy

Tools & Technologies: Git, Tailwind CSS, FastAPI, LLM

Soft Skills: Problem Solving, Analytical Thinking, Team Collaboration, Time Management